Case Studies in Operations Management

Mathematical Model

Business and Logistics Plan for "Fresh Veggie" (FV)

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1 Introduction

2 Problem Description

3 Model Assumptions

The following assumptions are adopted to simplify the modeling of the Fresh Veggie (FV) logistics network and to ensure tractability of the optimization problem.

- Demand for each product is assumed to be deterministic and constant over time.
- Households are assumed to be homogeneous in terms of consumption rates and ordering behavior.
- Each household is assumed to place exactly one order per week. This assumption approximates the stochastic ordering behavior described in the problem statement and allows for a tractable aggregate demand representation.
- All demand from the households served by the network is assumed to be fully satisfied. Service level constraints are not explicitly modeled.
- Customer behavior is assumed to be stable over time, with no cancellations or variability in ordering patterns.
- The coverage associated with each facility configuration $z \in \mathcal{Z}$ is assumed to be known and deterministic. The parameter $p(z)$ represents the number of households covered by configuration z and is precomputed.
- Households are assumed to be uniformly distributed within the service region.
- Each household is assigned to a single facility, and this assignment does not change over time.
- Coverage is not modeled geometrically within the optimization problem. Instead, it is incorporated through precomputed parameters.
- Congestion effects and routing complexities are not explicitly modeled. Transportation distances are approximated using average distances.
- All transportation and replenishment processes are assumed to be instantaneous. Lead times do not affect operational decisions and are only indirectly reflected through cost parameters.
- Transportation costs are assumed to be linear with respect to both distance and quantity transported.
- No capacity constraints are imposed on transportation modes (e.g., trucks, ships, or aircraft).
- Products are treated as continuous flows (in kilograms), and indivisibilities are ignored.
- No product losses occur during transportation or handling.
- Suppliers are assumed to have unlimited production capacity within their availability periods.
- Purchase prices and selling prices are assumed to be constant over time.

- Products are modeled independently, and no substitution effects between products are considered.
- There are no capacity constraints at ports, airports, or wholesale markets.
- All costs are modeled as linear functions. No economies of scale are considered in purchasing, transportation, or distribution.
- Transportation and delivery costs are approximated using average distances and cost coefficients.
- Inventory decisions are not explicitly modeled. All products are assumed to be immediately distributed upon arrival at the facilities.
- Product perishability and shelf-life constraints are not explicitly modeled in this formulation and are considered as part of future extensions of the model.

4 Sets

- CD: set of candidate distribution centers.
- DS: set of candidate dark stores.
- $MDCP$: set of candidate parking locations for mobile distribution centers.
- $F = CD \cup DS \cup MDCP$: set of all candidate facilities in the network.
- P: set of ports:
$$P = \{\text{San Antonio}, \text{Valparaíso}\}$$
- A: set of airports:
$$A = \{\text{Arturo Merino Benítez International Airport}\}$$
- WM: set of wholesale markets:
$$WM = \{\text{La Vega Central}, \text{Lo Valledor}\}$$
- S: set of supplier countries:
$$\begin{aligned} S = \{&\text{US}, \text{Mexico}, \text{Peru}, \text{Ecuador}, \\ &\text{Bolivia}, \text{Argentina}, \text{Spain}, \\ &\text{Chile-Scl}, \text{Chile-Vap}, \\ &\text{Chile-Maul}, \text{Chile-Coq}\} \end{aligned}$$
- $S_{cl} \subseteq S$: set of domestic suppliers located in Chile.
$$S = \{\text{Chile-Scl}, \text{Chile-Vap}, \text{Chile-Maul}, \text{Chile-Coq}\}$$
- K: set of products:
$$K = \{\text{Roma Tomatoes}, \text{Cauliflower}, \text{Broccoli}, \text{Asparagus}, \text{Green Bell Pepper}\}$$
- T: set of weeks in the planning horizon:
$$T = \{1, 2, \ldots, 52\}$$
- R: set of transportation modes:
$$R = \{\text{sea}, \text{air}, \text{truck}\}$$
- $\mathcal{Z}$: set of all feasible facility configurations, where each $z \in \mathcal{Z}$ is a binary vector in $\{0, 1\}^{|F|}$, excluding the null configuration.
- $I \subseteq \mathcal{Z}$: set of singleton facility configurations, where each $i \in I$ is a binary vector with exactly one component equal to 1 and all others equal to 0.

5 Parameters

5.1 Distance Parameters

All distances are measured in kilometers (km).

- $d_{ij}^{CD,DS}$: distance from distribution center $i \in CD$ to dark store $j \in DS$.
- d_{ij}^{MDCP}: distance between mobile distribution center parking locations $i, j \in MDCP$.
- $d_{ij}^{CD,P}$: distance from distribution center $i \in CD$ to port $j \in P$.
- $d_{ij}^{CD,A}$: distance from distribution center $i \in CD$ to airport $j \in A$.
- $d_{ij}^{CD,WM}$: distance from distribution center $i \in CD$ to wholesale market $j \in WM$.
- $d_{ij}^{S_{cl},CD}$: distance from Chilean supplier $i \in S_{cl}$ to distribution center $j \in CD$ (direct transport).
- $\bar{d}_f$: average distance (km) from facility $f \in F$ to the households it serves.

5.2 Coverage Parameters

- $z \in \{0,1\}^{|F|}$: binary vector representing a facility configuration, where each component indicates whether a specific facility is open (1) or closed (0). The components of z are ordered such that the first $|CD|$ elements correspond to distribution centers (CD), the next $|DS|$ elements correspond to dark stores (DS), and the last $|MDCP|$ elements correspond to mobile distribution centers (MDCP).
- $p(z)$: number of households that can be covered under facility configuration z, for all $z \in \mathcal{Z}$.
- $P_i = p(i)$: maximum number of households that can be covered by singleton configuration $i \in I$.

5.3 Configuration Parameters

- $\delta_{i,z}^{CD}$: binary parameter equal to 1 if distribution center $i \in CD$ is active in configuration $z \in \mathcal{Z}$, and 0 otherwise.
- $\delta_{j,z}^{DS}$: binary parameter equal to 1 if dark store $j \in DS$ is active in configuration $z \in \mathcal{Z}$, and 0 otherwise.
- $\delta_{f,z}$: binary parameter equal to 1 if facility $f \in F$ is active in configuration $z \in \mathcal{Z}$, and 0 otherwise.

5.4 Cost Parameters

- $c^{buy}_{k,s,r}$: purchase cost (USD/kg) of product $k \in K$ from supplier $s \in S$ using transport mode $r \in R$.
- $c^{WM}_{k,w}$: purchase cost (USD/kg) of product $k \in K$ at wholesale market $w \in WM$.
- v_k: selling price (USD/kg) of product $k \in K$.
- c^{tr}: transportation cost per kg per km by truck (USD/kg/km).
- c^{del}: last-mile delivery cost per kg per km (USD/kg/km).
- c^{mov}: movement cost per kilometer (USD/km) for the mobile distribution center.

5.5 Availability Parameters

- $a_{k,s,t}$: binary parameter equal to 1 if product $k \in K$ is available from supplier $s \in S$ in period $t \in T$.
- $g_{k,r}$: binary parameter equal to 1 if product $k \in K$ can be transported using mode $r \in R$.
- $u^P_{s,p}$: binary parameter equal to 1 if supplier $s \in S$ can ship through port $p \in P$.
- $u^A_{s,a}$: binary parameter equal to 1 if supplier $s \in S$ can ship through airport $a \in A$.

5.6 Transportation Parameters

- $t^{first}_{s,p,r}$: transportation time (hours) from supplier $s \in S$ to entry point $p \in P \cup A$ using mode $r \in R$. For domestic suppliers ($s \in$ Chile), this value is equal to 0.
- α^{LH}_r: long-haul travel time coefficient (hours/km) for transportation mode $r \in R$.
- $t^{LH}_{i,j,r}$: long-haul transportation time (hours) between locations i and j, computed as:

$$t^{LH}_{i,j,r} = \alpha^{LH}_r \cdot d_{i,j}$$

- α^{city}: urban travel time coefficient (hours/km) for local distribution.
- $t^{city}_{i,j}$: urban transportation time (hours) between facilities or to customers, computed as:

$$t^{city}_{i,j} = \alpha^{city} \cdot d_{i,j}$$

- $g(z)$: minimum total travel distance (km) required for the mobile distribution center to visit all active parking locations in configuration $z \in \mathcal{Z}$ and return to its starting point.

5.7 Demand Parameters

- q_k: demand rate (kg/person/week) of product $k \in K$.
- $\bar{h}$: average number of people per household (persons/household).
- f^{del}: flat delivery fee per order (USD/order), equal to 2.5 USD.

6 Decision Variables

- $x_z^{config} \in \{0,1\}$: equals 1 if facility configuration $z \in \mathcal{Z}$ is selected, and 0 otherwise.
- $x_{k,s,r,t}^{sup} \geq 0$: quantity (kg) of product $k \in K$ purchased from supplier $s \in S$ using transportation mode $r \in R$ in period $t \in T$.
- $x_{k,w,t}^{WM} \geq 0$: quantity (kg) of product $k \in K$ purchased from wholesale market $w \in WM$ in period $t \in T$.
- $y_{k,p,i,t}^{P} \geq 0$: quantity (kg) of product $k \in K$ transported from port $p \in P$ to distribution center $i \in CD$ in period $t \in T$.
- $y_{k,a,i,t}^{A} \geq 0$: quantity (kg) of product $k \in K$ transported from airport $a \in A$ to distribution center $i \in CD$ in period $t \in T$.
- $y_{k,s,i,t}^{S} \geq 0$: quantity (kg) of product $k \in K$ transported from Chilean supplier $s \in S_{cl}$ to distribution center $i \in CD$ in period $t \in T$.
- $y_{k,w,i,t}^{WM} \geq 0$: quantity (kg) of product $k \in K$ transported from wholesale market $w \in WM$ to distribution center $i \in CD$ in period $t \in T$.
- $y_{k,i,j,t}^{CD,DS} \geq 0$: quantity (kg) of product $k \in K$ transported from distribution center $i \in CD$ to dark store $j \in DS$ in period $t \in T$.
- $h_{f,t} \geq 0$: number of households served by facility $f \in F$ in period $t \in T$.

7 Objective Function

The objective is to maximize total profit over the planning horizon, defined as total revenue minus all relevant costs.

Revenue:

$$\sum_{t \in T} \left[\sum_{z \in \mathcal{Z}} \left(p(z) \cdot x_z^{config} \right) \cdot \left(\bar{h} \cdot \sum_{k \in K} q_k \cdot v_k + f^{del} \right) \right]$$

Purchasing Cost:

$$\sum_{t \in T} \left[\sum_{k \in K} \sum_{s \in S} \sum_{r \in R} c_{k,s,r}^{buy} \, x_{k,s,r,t}^{sup} + \sum_{k \in K} \sum_{w \in WM} c_{k,w}^{WM} \, x_{k,w,t}^{WM} \right]$$

Long Haul Cost:

$$\sum_{t \in T} \left[\sum_{k \in K} \sum_{p \in P} \sum_{i \in CD} c^{tr} \cdot d_{i,p}^{CD,P} \cdot y_{k,p,i,t}^{P} + \sum_{k \in K} \sum_{a \in A} \sum_{i \in CD} c^{tr} \cdot d_{i,a}^{CD,A} \cdot y_{k,a,i,t}^{A} \right.$$

$$\left. + \sum_{k \in K} \sum_{s \in S_{cl}} \sum_{i \in CD} c^{tr} \cdot d_{s,i}^{S_{cl},CD} \cdot y_{k,s,i,t}^{S} + \sum_{k \in K} \sum_{w \in WM} \sum_{i \in CD} c^{tr} \cdot d_{i,w}^{CD,WM} \cdot y_{k,w,i,t}^{WM} \right]$$

Inter-Facility Transportation Cost:

$$\sum_{t \in T} \left[\sum_{k \in K} \sum_{i \in CD} \sum_{j \in DS} c^{tr} \cdot d_{i,j}^{CD,DS} \cdot y_{k,i,j,t}^{CD,DS} \right]$$

Last-Mile Delivery Cost:

$$\sum_{t \in T} \sum_{f \in F} c^{del} \, \bar{d}_f \, h_{f,t} \left(\bar{h} \sum_{k \in K} q_k \right)$$

Mobile Distribution Center Movement Cost:

$$|T| \cdot c^{mov} \sum_{z \in \mathcal{Z}} g(z) \, x_z^{config}$$

8 Constraints

This section presents the core constraints required to ensure consistency between the selected facility configuration, the use of the logistics network, and the households served in each period.

8.1 Configuration Selection

The model must select exactly one facility configuration from the set of feasible configurations.

$$\sum_{z \in \mathcal{Z}} x_z^{config} = 1 \qquad \forall$$

This constraint ensures that the network design is uniquely defined by a single configuration $z \in \mathcal{Z}$.

8.2 Activation of Distribution Centers for Inbound Flows

A distribution center can receive inbound product flows only if it is active in the selected configuration.

$$\sum_{k \in K} \left(\sum_{p \in P} y^{P}_{k,p,i,t} + \sum_{a \in A} y^{A}_{k,a,i,t} + \sum_{s \in S_{cl}} y^{S}_{k,s,i,t} + \sum_{w \in WM} y^{WM}_{k,w,i,t} \right) \leq M^{CD,in}_{i,t} \sum_{z \in \mathcal{Z}} \delta^{CD}_{i,z} \, x_z^{config} \qquad \forall i \in CD,\ t \in T$$

This constraint prevents products from entering a distribution center that is not part of the chosen network configuration.

8.3 Activation of Distribution Centers for Outbound Flows to Dark Stores

A distribution center can send products to dark stores only if it is active in the selected configuration.

$$\sum_{k \in K} \sum_{j \in DS} y^{CD,DS}_{k,i,j,t} \leq M^{CD,out}_{i,t} \sum_{z \in \mathcal{Z}} \delta^{CD}_{i,z} \, x_z^{config} \qquad \forall i \in CD,\ t \in T$$

This ensures that outbound inter-facility flows can only originate from active distribution centers.

8.4 Activation of Dark Stores for Inbound Flows

A dark store can receive products from distribution centers only if it is active in the selected configuration.

$$\sum_{k \in K} \sum_{i \in CD} y^{CD,DS}_{k,i,j,t} \leq M^{DS}_{j,t} \sum_{z \in \mathcal{Z}} \delta^{DS}_{j,z} \, x_z^{config} \qquad \forall j \in DS,\ t \in T$$

This constraint guarantees that dark stores not selected in the chosen configuration cannot be used as logistics nodes.

8.5 Activation of Customer Service by Facility

A facility can serve households only if it is active in the selected configuration.

$$h_{f,t} \leq M^H_{f,t} \sum_{z \in \mathcal{Z}} \delta_{f,z}\, x_z^{config} \qquad \forall f \in F,\ t \in T$$

This condition links customer service decisions to the selected network structure.

8.6 Consistency Between Covered Households and Households Served

The total number of households served in each period must match the number of households covered by the selected configuration.

$$\sum_{f \in F} h_{f,t} = \sum_{z \in \mathcal{Z}} p(z)\, x_z^{config} \qquad \forall t \in T$$

This constraint ensures consistency between the strategic coverage decision and the operational allocation of households across facilities.

8.7 Non-Negativity and Binary Restrictions

The facility configuration variable is binary, while all flow and service variables are non-negative.

$$x_z^{config} \in \{0, 1\} \qquad \forall z \in \mathcal{Z}$$

$$x^{sup}_{k,s,r,t} \geq 0 \qquad \forall k \in K,\ s \in S,\ r \in R,\ t \in T$$

$$x^{WM}_{k,w,t} \geq 0 \qquad \forall k \in K,\ w \in WM,\ t \in T$$

$$y^{P}_{k,p,i,t} \geq 0 \qquad \forall k \in K,\ p \in P,\ i \in CD,\ t \in T$$

$$y^{A}_{k,a,i,t} \geq 0 \qquad \forall k \in K,\ a \in A,\ i \in CD,\ t \in T$$

$$y^{S}_{k,s,i,t} \geq 0 \qquad \forall k \in K,\ s \in S_{cl},\ i \in CD,\ t \in T$$

$$y^{WM}_{k,w,i,t} \geq 0 \qquad \forall k \in K,\ w \in WM,\ i \in CD,\ t \in T$$

$$y^{CD,DS}_{k,i,j,t} \geq 0 \qquad \forall k \in K,\ i \in CD,\ j \in DS,\ t \in T$$

$$h_{f,t} \geq 0 \qquad \forall f \in F,\ t \in T$$

9 Extensions and Special Considerations

10 Data and Calibration

11 Computational Implementation

12 Results

13 Analysis and Discussion

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